Experiment I: Millikan Oil Drop Experiment

I. Preparation Guide

1. The experimental principle of the Millikan oil drop method for measuring the electron charge;
2. Derive the calculation formula for the charge of the oil drop in this experiment;
3. Draw a schematic diagram of the experimental device structure;
4. How to qualitatively determine the volume and charge of the oil drop?

II. Objectives and Tasks

1. Observe the motion of charged oil droplets in gravitational and electric fields;
2. Select appropriate oil droplets and use the static method to determine their charge values;
3. Select appropriate oil droplets and use the dynamic method to determine their charge values (Optional);
4. Calculate the electron charge value e, compare it with the recognized value, and draw appropriate conclusions.

III. Principle

When oil droplets are sprayed out through a sprayer, the tiny oil droplets will be charged due to friction. The charge q is an integer multiple of the elementary charge (the absolute value of the electron charge, the charge of the proton) e. Its value is determined by observing and measuring the movement of the charged oil droplets in an electric field.

1. Static Method for Measuring the Charge of Oil Droplets

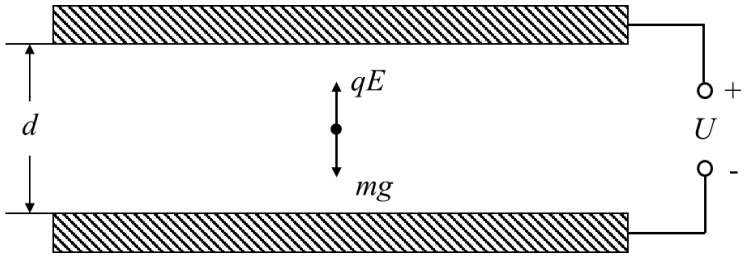


Figure 1-1 Charged Oil Droplets in Parallel Plates

Place an oil droplet with a mass of m and a charge of q between parallel plates with a voltage of U_n and a spacing of d, as shown in Figure 1-1. The oil droplet will be affected by both gravity and electric force (the air buoyancy is very small compared to the electric force and gravity and can be ignored). Choose the appropriate voltage polarity and voltage value to balance the electric force on the oil droplet with the gravity, that is.

$$mg = \frac{q U_n}{d} \quad (1\text{-}1)$$

The oil droplet will be suspended in the electric field and maintain equilibrium. Therefore, by measuring the mass m of the oil droplet, the equilibrium voltage U_n, and the distance d between the parallel plates, the charge q of the oil droplet can be obtained from (1-2).

$$q = \frac{mgd}{U_n} \quad (1\text{-}2)$$

Since the charge q carried by the oil droplet is an integer multiple of the elementary charge e, that is, $q = ne$ (n=±1, ±2, ...), for the same oil droplet, if it carries a charge of q_1, q_2, q_3, ..., then its equilibrium voltage U_{n1}, U_{n2}, U_{n3}, ... measured by the experiment can only be some discontinuous specific values (discrete).

In the experiment, we measured the various charge values q_1, q_2, q_3, ..., and then calculated their greatest common divisor, which is the elementary charge e.

In the experiment, the mass m of the oil droplet is on the order of 10^{-15}kg, which is extremely difficult to measure directly, so it is necessary to obtain it through the following indirect measurement method.

Under the action of surface tension, the oil droplet is generally spherical, and its mass can be expressed as

$$m = \frac{4}{3}\pi\rho(a)^{3} \quad (1\text{-}3)$$

In the formula, ρ is the density of the oil, and a is the radius of the oil droplet. Under the premise of knowing ρ, a is measured, and the mass m of the oil droplet can be obtained by formula (1-3). The radius a of the oil droplet can be measured in the following way.

If the voltage on the parallel plates is removed, so that the electric field force on the oil droplet is zero, the oil droplet will fall. In the process of falling, the oil droplet will be affected by gravity $G=mg$, air buoyancy $F_1 = \frac{4}{3}\pi\rho' g(a)^3$ and air viscous friction resistance F_2, where ρ' is the density of air. According to Stokes' law,

$$F_2 = 6\pi a\eta v \quad (1\text{-}4)$$

Where v is the speed of the oil droplet falling, and η is the viscosity of the air.

Since the viscous friction resistance of the air to the oil droplet is proportional to the falling speed of the oil droplet, the oil droplet initially accelerates and falls. As the falling speed increases, the viscous friction resistance also increases, and the acceleration gradually decreases. When the falling speed increases to a certain value v_s, the external force on the oil droplet is balanced, that is, $G=F_1+F_2$. At this time, the oil droplet falls at a uniform speed, that is

$$\frac{4}{3}\pi\rho g(a)^{3} = \frac{4}{3}\pi\rho' g(a)^{3} + 6\pi a\eta v_s \quad (1\text{-}5)$$

From the above formula, we can get the radius of the oil droplet

$$a =\sqrt{\frac{9\eta v_s}{2g(\rho-\rho')}} \quad (1\text{-}6)$$

Since $\rho>>\rho'$, that is, the buoyancy of air on oil droplets can be ignored, the above formula can be simplified to

$$a =\sqrt{\frac{9\eta v_s}{2g\rho}} \quad (1\text{-}7)$$

On the other hand, since the oil droplets are very small and their diameter is smaller than the gap between air molecules, the air cannot be regarded as a continuous and uniform medium. Therefore, Stokes' law needs to be corrected and the viscous friction resistance of air to the oil droplets is corrected to

$$F_2 =\frac{6\pi a\eta v_s}{1 +\frac{b}{pa}} \quad (1\text{-}8)$$

Where p is the atmospheric pressure in Pa, the correction constant b=8.22×10^{-3}$m\cdot Pa$, and the corrected oil droplet radius expression is

$$a =\sqrt{\frac{9\eta v_s}{2g\rho\ (1 +\frac{b}{pa})}} \quad (1\text{-}9)$$

The unknown parameter a on the right side of the above formula is in the correction term (1+b/pa) and does not need to be very accurate, so its value is still calculated using formula (1-7).

Substituting formula (1-9) into formula (1-3), the expression for the mass of the oil droplet is

$$m =\frac{4\pi}{3}\rho\ (\frac{9\eta v_s}{2g\rho\ (1 +\frac{b}{pa})})^{\frac{3}{2}} \quad (1\text{-}10)$$

Substituting the above equation into equation (1-2), we get the expression of the charge carried by the oil droplet

$$q =\frac{18\pi}{\sqrt{2g\rho}} \cdot \frac{d}{U_n} \left(\frac{\eta\ v_s}{1 +\frac{b}{pa}}\right)^{\frac{3}{2}} \quad (1\text{-}11)$$

In the above formula, it is also necessary to measure the speed v_s of the oil drop falling at a uniform speed, which can be determined by measuring the time t taken by the oil drop to fall at a certain distance l at a uniform speed, that is

$$v_s =\frac{l}{t} \quad (1\text{-}12)$$

Then formula (1-7) becomes

$$a =\sqrt{\frac{9\eta\ l}{2g\rho\ t}} \quad (1\text{-}13)$$

Substituting equations (1-12) and (1-13) into equation (1-11), we get the expression for the charge carried by the oil droplet

$$q =\frac{18\pi}{\sqrt{2g\rho}} \cdot \frac{d}{U_n} \left(\frac{\eta\ l}{t \left(1 + \frac{b}{p} \sqrt{\frac{2g\rho\ t}{9\eta\ l}} \right)} \right)^{\frac{3}{2}} \quad (1\text{-}14)$$

Under the premise that the density of oil ρ, gravitational acceleration g, viscosity coefficient of air η, atmospheric pressure p, correction constant b, distance between parallel plates d, and the distance l that the oil drop falls at a uniform speed are known, it is only necessary to measure the equilibrium voltage U_n, then

remove the voltage and let the oil drop fall freely in the air. It only takes a very short time for the oil drop to reach a uniform fall. By measuring the time t it takes to fall a given distance l, the charge q can be calculated.

2. Dynamic Method for Measuring the Charge of Oil Droplets

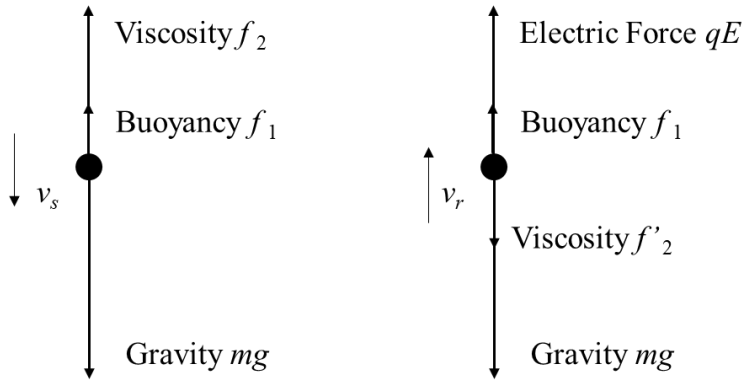


Figure 1-2 The left picture shows the force on an oil droplet falling in a gravity field, and the right picture shows the force on an oil droplet rising in an electric field.

In the above experiment, we used the "static method" to measure the amount of electron charge. Corresponding to this is a "dynamic method", that is, after finding a suitable oil drop and determining the equilibrium voltage, the oil drop can be lifted above the 0 scale line, and then the working voltage is removed to make the oil drop fall, and the time t_1 taken for the oil drop to fall a given distance l is recorded, and then the voltage is increased to make the oil drop rise, and the time t_2 taken to rise a given distance l and the rising working voltage U are recorded.

In the process of oil drop falling, according to Stokes' law, the resistance of the viscous fluid (here is air) to the spherical moving object is proportional to the object's speed, and its proportional coefficient is $6\pi\eta a$, where η is the air viscosity, a is the object radius, ρ and ρ' are the oil drop and air densities respectively. Assuming that the oil drop falls at a uniform speed of speed vs, then there is

$$\frac{4\pi}{3} a^3 \rho g = \frac{4\pi}{3} a^3 \rho' g + 6\pi \eta a v_s \quad (1\text{-}15)$$

In the rising process of the oil drop, assuming that the oil drop rises at a constant speed v_r, then

$$\frac{4\pi}{3} a^3 \rho g + 6\pi \eta a v_r = \frac{4\pi}{3} a^3 \rho' g + qE \quad (1\text{-}16)$$

Since the density of oil droplets is much greater than that of air, the buoyancy of air on oil droplets can be ignored, so q is

$$q = \frac{4\pi a^3 (\rho - \rho')g}{3Ev_s}(v_s + v_r) \approx \frac{4\pi a^3 \rho gd}{3Uv_s}(v_s + v_r) \quad (1\text{-}17)$$

Ignoring buoyancy, we can obtain from formula (1-15)

$$v_s = \frac{2ga^2}{9\eta}\rho \quad (1\text{-}18)$$

Therefore

$$a = \sqrt{\frac{9\eta v_s}{2\rho g}} \quad (1\text{-}19)$$

The time taken for the oil drop to fall at a constant speed for a distance of l is t_1, and the time taken for an oil drop to rise at a constant speed for a distance of l is t_2, so v_s and v_r are v_s =l/ t_1, v_r =l/ t_2 respectively.

Considering that the diameter of the oil drop is equivalent to the gap between air molecules, the air can no longer be regarded as a continuous medium, and its air viscosity η needs to be corrected to η'

$$\eta' = \frac{\eta}{1 + \frac{b}{pa}} \tag{1-20}$$

Here p is the air pressure, b is the correction constant, b=0.00823N/m, so equation (1-19) can be corrected to

$$a = \sqrt{\frac{9\eta v_s}{2\rho g \left(1 + \frac{b}{pa} \right)}} \tag{1-21}$$

Since the radius a is also included in the correction term, when the accuracy requirement is not too high, the correction term a can be calculated by formula (1-19).

Substituting (1-20) into (1-17), we can obtain

$$\begin{aligned} q &= \frac{4\pi \left[\frac{9\eta v_s}{2\rho g \left(1 + \frac{b}{pa} \right)} \right]^{\frac{3}{2}} \rho g d(v_s + v_r)}{3Uv_s} \\ &= \frac{18\pi}{\sqrt{2\rho g}} \cdot \frac{d}{U} \cdot \left(1 + \frac{t_1}{t_2} \right) \left[\frac{\eta l}{t_1 \left(1 + \frac{b}{p} \sqrt{\frac{2\rho g t_1}{9\eta l}} \right)} \right]^{\frac{3}{2}} \end{aligned} \tag{1-22}$$

3. Calculation of Electron Charge

In the experiment, it is necessary to measure the charge carried by different oil droplets to verify the discontinuity of the charge and calculate the value of the elementary charge e. Due to the existence of experimental errors, it is difficult to accurately calculate the value of the elementary charge using the greatest common divisor method. Therefore, the following method can be used:

Use the recognized elementary charge value e=1.602×10^{-19}C to divide the charge q carried by the oil droplet measured in the experiment. If the experiment is successful, then q/e will be very close to a certain integer value n. It can be considered that n is the number of elementary charges (or electrons) carried by the oil droplet, and q/n is the elementary charge value measured in the experiment.

It should be noted that for the same oil droplet, multiple measurements (5 times) must be performed patiently to avoid the influence of natural fluctuations.

IV. Operation

1. Instrument Adjustment

(1) Horizontal Adjustment

Adjust the leveling screw knobs of the main unit of the experimental instrument (when looking down, turn the knobs clockwise to lower the platform and counterclockwise to raise the platform) until the level bubble is exactly in the center (Note: it is strictly forbidden to turn the knob on the level bubble). Level the experimental platform so that the direction of the equilibrium electric field is parallel to the direction of gravity to avoid experimental errors. Whether the plate plane is horizontal determines whether the oil droplets drift left and right during the falling or lifting process.

(2) Sprayer Adjustment

Slowly pour a small amount of watch oil into the oil storage chamber of the sprayer so that the watch oil submerges the bottom of the oil lifting pipe. Do not pour too much oil to avoid accidentally pouring oil into the oil drop box and clogging the oil drop hole during the experiment. Stand the sprayer upright and squeeze the air bag by hand to fill the oil lifting pipe with watch oil.

(3) Connect the Experimental Instrument

(a) Turn on the power of the experimental instrument and the power of the LCD monitor.

(b) Press any key on the host, the monitor will display the parameter setting interface. First, set the experimental method, and then set the gravity acceleration, oil density, atmospheric pressure, and oil drop distance according to the local environment. "←" indicates the left shift key, "→" indicates the right shift key, and "+" indicates the data setting key.

(c) After pressing the "确认"(Confirm) key, the experimental interface will appear, the "计时"(Timing) "开始/结束"(Start/End) key should be set to "结束"(End), the "0V/工作"(0V/Work) key should be set to "0V"(0V), and the "平衡/提升"(Balance/Lift) key should be set to "平衡"(Balance).

(4) CCD Imaging System Adjustment

Turn on the oil inlet switch and spray oil mist from the spray port. At this time, a large number of moving oil droplets should appear on the monitor. If no oil droplet image is seen, adjust the focus knob or check whether the sprayer is working properly.

2. Select Appropriate Oil Droplets

If the oil droplets are too small, the influence of Brownian motion will be very strong, the equilibrium voltage will not be easy to adjust, and the time error will also increase; if the oil droplets are too large and fall too fast, the relative time error will increase, and the probability of the oil droplets carrying multiple electrons will increase. The appropriate oil droplets should preferably carry less than 10 electrons.

General operation method: Press the "0V/工作"(0V/Work) key to adjust the initial working voltage to about 150~250V. Spray the oil droplets and adjust the focus knob to display most of the oil droplets on the screen. It can be seen that the charged oil droplets quickly rise out of the field of view, and the uncharged oil droplets fall out of the field of view. After about 10s, the oil droplets decrease. Select the oil droplet that rises slowly as the temporary target oil droplet, switch the "0V/工作"(0V/Work) key, then the voltage between the plates is 0V, select the one with a falling speed of 0.2~0.5 grid/s as the final target oil droplet among the temporary target oil droplets, and adjust the focus knob to make the oil droplet the smallest and brightest.

In order to more quickly screen out oil droplets with appropriate charge, the initial working voltage can also be set to 350~400V, so that oil droplets with more charge will rise and move out of the field of view more quickly, and the oil droplets in the field of view will decrease rapidly. Find oil droplets with slower rising speed as the target, and gradually reduce the value of the working voltage (reduced to 150~250V) on this basis until the rising speed of the oil droplets slows down and reaches a balanced state.

3. Confirmation of Equilibrium Voltage

After focusing the target oil droplet to the smallest and brightest, carefully adjust the "电压调节" (Voltage Adjustment) to balance the oil droplet on a certain grid line, wait for a while (about two minutes), and observe whether the oil droplet floats away from the grid line. If the oil droplet always floats away in the same direction, the equilibrium voltage needs to be readjusted; if it is basically stable on the grid line or only makes slight Brownian motion above and below the grid line, it can be considered that the oil droplet has reached mechanical equilibrium, and the voltage at this time is the equilibrium voltage.

4. Static Method to Measure the Charge of Oil Droplets

Balance the oil droplet on the first grid line at the top of the screen, switch the working state button to "0V", and the green indicator light will light up. At this time, the upper and lower plates are grounded at the same time, the electric field force is zero, and the oil droplet falls under the action of gravity, buoyancy and air resistance. The oil droplet first undergoes a period of variable speed motion and then changes to uniform speed motion, but the time of variable speed motion is very short (less than 0.01s, which is equivalent to the accuracy of the timer), so it can be considered that the oil droplet falls immediately at a uniform speed. When the oil droplet falls to the grid line with a "0" mark, immediately press the "计时" (Timing) button, and the timer starts to record the time of the oil droplet falling; when the oil droplet falls to the grid line with a distance mark (1.6), press the "计时" (Timing) button again, the timer stops timing, and the oil droplet stops falling. The "0V/工作" (0V/Work) button automatically switches to "工作" (Work), and the "平衡/提升"(Balance/Lift) button is in "平衡"(Balance). The measured data of this time can be recorded on the screen by pressing the "确认"(Confirm) button. Switch the "平衡/提升"(Balance/Lift) button to "提升"(Lift). At this time, the plate voltage increases by about 200V on the basis of the original balance voltage, and the oil droplet immediately moves upward. When the oil droplet is lifted to the top of the screen, switch to "平衡"(Balance), find the balance voltage, and perform the next measurement. A total of 3 different oil droplets were measured, and each oil droplet was measured 5 times at least.

5. Dynamic Method to Measure the Charge of Oil Droplets (Optional)

After finding a suitable oil droplet using the above method, measure the droplet's falling time t_1, rising time t_2, and rising voltage U and record them. A total of 3 different oil droplets were measured, and each oil droplet was measured 5 times at least.

V. Precautions

1. The mechanical position of the CCD box, screws, and camera lens cannot be changed, otherwise it will affect the image distance and imaging angle.

2. Pay attention to adjusting the oil inlet switch to avoid the influence of external air flow on oil droplet measurement.

3. There is high voltage in the instrument, so avoid touching the electrodes with your hands.

4. Before the experiment, the two plates must be adjusted to be parallel to ensure that the electric field is vertical and parallel to gravity;

5. It is recommended to select an oil droplet with a balance voltage of about 150~250V and a drop time of 16-40s for 1.6mm. Such an oil droplet has a moderate charge and is suitable for experimental measurement.

Related Reference Data Used in the Experiment

Oil Density: $\rho = 981\ \mathrm{kg \cdot m^{-3}}$

Gravity Acceleration: $g = 9.78\ \mathrm{m \cdot s^{-2}}$

Air Viscosity Coefficient: $\eta = 1.83\times10^{-5}\cdot\mathrm{kg \cdot m^{-1} \cdot s^{-1}}$

Oil Droplet Uniform Descent Distance: $l = 1.60\times10^{-3}\mathrm{m}$

Correction Constant: $b = 8.22\times10^{-3}\mathrm{m \cdot Pa}$

Atmospheric Pressure (Shenzhen): $P = 1.0098\times10^{5}\ \mathrm{Pa}$

Parallel Plate Distance: $d = 5.00\times10^{-3}\mathrm{m}$

VI. Requirements for Experimental Reports

1. For the static method, record the equilibrium voltage U_n and the falling time t; for the dynamic method (optional), record the rising voltage U, the falling time t_1, and the rising time t_2.
2. Use the static method and the dynamic method (dynamic method is optional) to measure at least 3 oil droplets respectively, record the charge q_i of each oil droplet, calculate " q_i /e", and round off the quotient to get the number of electrons n_i carried by each oil droplet; then get the basic charge e_i for each measurement, and then calculate e for n times measurements, and compare it with the recognized elementary charge value $e = 1.602\times10^{-19}$ C to calculate the percentage error.

VI. Questions

1. When track and observe a certain oil drop, the original clear image becomes blurred. What may be the reason?
2. Due to the volatilization of oil, the quality of the oil drop will continue to decline. When track and measure the same oil drop for a long time, what measurement quantities will change due to the volatilization of the oil drop?